

FEVEN ABEBE BEJIGA

Qiyas Trainee — Data Science and AI Engineering

1. PERSONAL INFORMATION

Full Name: Feven Abebe Bejiga

Phone: +251 988 997 925

Email: fevenabebe777@gmail.com

Location: Addis Ababa, Ethiopia

Qiyas Student ID: qiyas-2026-003641

LinkedIn / Portfolio: [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Primary Career Goal: Employment / Internship

Availability: Immediately

2. PROFESSIONAL SUMMARY

AI/ML and data science professional with a BSc in Health Informatics and 3+ years of experience in digital health systems and data management, now pursuing an MSc in Artificial Intelligence at Addis Ababa University. Completed Qiyas Data Science and AI Engineering training, building hands-on skills in machine learning, deep learning, NLP, and model deployment. Co-author of a federated learning paper submitted to PanAfriCon AI 2026. Comfortable across the ML lifecycle — from model development (PyTorch, TensorFlow, Scikit-learn) through deployment (FastAPI, Docker, CI/CD) — and seeking opportunities to apply AI and data science skills to real-world problems.

3. EDUCATION

- MSc in Artificial Intelligence — Addis Ababa University (AAiT), Nov 2024 – Present
- BSc in Health Informatics — University of Gondar, 2022 (CGPA 3.86)

4. QIYAS TRAINING

Training Category / Program: Data Science and AI Engineering

Training Period: May 2026 – September 2026 (Completed)

Key Competencies:

- Python programming, SQL, and data manipulation
- Exploratory Data Analysis (EDA), data preprocessing, data visualization, and statistical analysis
- Machine learning, deep learning, and Natural Language Processing (NLP), including sentiment analysis
- Model development, evaluation, and deployment for practical AI/data science applications
- Time series analysis and forecasting, and web scraping

5. WORK EXPERIENCE / INTERNSHIP

Experience 1

Organization / Employer: CDHi, University of Gondar

Role / Position: Digital Health Developer

Period: Jul 2023 – Jan 2026

Responsibilities / Achievements:

- Customized EMR and DHIS2 systems to align with hospital workflows and organizational requirements, working closely with clinical and administrative staff to translate their needs into system configurations.
- Implemented the WHO Digital Adaptation Toolkit (DAK) within the University of Gondar Referral Hospital's EMR environment, helping bring the hospital's electronic records in line with national digital health standards.
- Managed and maintained MySQL databases underlying the hospital's digital health systems, including data integrity checks and routine database administration.
- Generated reports from EMR and DHIS2 data to support hospital decision-making and reporting requirements, alongside core development work.

- Developed DHIS2 Tracker tools to support Health and Demographic Survey activities, enabling structured collection and management of survey data.
- Built aggregate data-collection tools for ISHO and supported the analysis of the resulting health information.
- Customized an open-source project management system to fit CDHI's day-to-day operational requirements.

Experience 2

Organization / Employer: Center for Digital Health and Implementation (CDHI)

Role / Position: Fellow

Period: Nov 2022 – Jun 2023

Responsibilities / Achievements:

- Supported digital health and health information systems activities.
- Built practical experience in digital health technologies and health information management.

6. TECHNICAL & PRACTICAL SKILLS

- Python, SQL, C++, Java, JavaScript
- Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers
- NLP & RAG: Natural Language Processing, Hugging Face Transformers, Retrieval-Augmented Generation
- Computer Vision: Image Processing, Image Classification, Object Detection, OpenCV
- Data Science: pandas, NumPy, SciPy, Matplotlib, EDA, Statistical Analysis, Time Series Analysis
- Engineering & Deployment: FastAPI, REST APIs, Docker, Git, GitHub Actions, Streamlit
- Digital Health: EMR, DHIS2, Health Information Systems

7. SELECTED PROJECTS

AI-Driven Agricultural Decision Support App — *Python · AI/ML · Image Classification · RAG · Edge AI · Chatbot*

- Building an offline-first AI decision-support concept for agricultural extension workers, combining on-device crop-disease image classification for maize and wheat with a multilingual RAG knowledge base.
- Built a conversational chatbot interface so farmers and extension workers can ask questions in natural language and get grounded, easy-to-understand answers.
- Designed a human-in-the-loop escalation approach for uncertain cases, backed by an early RAG proof of concept.

Federated Learning with Metaheuristic Client Selection — *Python · PyTorch · FedProx · GA / PSO / SA*

- Built a federated learning framework for non-IID client data using PyTorch and FedProx, and designed a multi-client experimental pipeline covering local training, global aggregation, and iterative communication rounds.
- Investigated Genetic Algorithm, Particle Swarm Optimization, and Simulated Annealing as intelligent client-selection strategies, evaluated against a random baseline on training loss and classification accuracy.

ROS2 Autonomous Lane-Following Robot — *ROS2 Humble · Gazebo · Python · Computer Vision · PID Control*

- Developed a lane-following robot simulation using HSV-based color segmentation for lane detection and a ROS2 node to compute lane error.
- Implemented PID-based steering control, integrating perception and control into a single autonomous robotics pipeline.

House Price Prediction System — *Python · Scikit-learn · CatBoost · FastAPI · Streamlit · Docker*

- Built an end-to-end ML system for real-estate price prediction: data preprocessing and feature engineering, a FastAPI REST API for serving predictions, and a Streamlit front end.
- Containerized the application with Docker and implemented automated testing with GitHub Actions CI.

Fraud Detection Machine Learning Project — *Python · Scikit-learn · Random Forest · Logistic Regression*

- Developed and compared Logistic Regression and Random Forest classifiers for highly imbalanced fraud-detection datasets, using F1-score and AUC-PR as evaluation metrics appropriate to the class imbalance.

8. PUBLICATIONS

- “Lightweight Metaheuristic Optimization for Client Selection in Resource-Constrained Federated Learning Systems” — co-authored with Zehara Yassin and Beakal Gizachew (Addis Ababa University); submitted to PanAfriCon AI 2026.
- Compares Genetic Algorithm, Particle Swarm Optimization, and Simulated Annealing against random client selection on FEMNIST, CIFAR-10, and CIFAR-100 using FedProx (5 seeds, 100 rounds).

9. CERTIFICATIONS / ADDITIONAL TRAINING

- Data Science and AI Engineering — Qiyas, Addis Ababa University, 2026
- SAFEE KAIM — Kifiya AI Mastery Training Program, 10 Academy Women-Only Cohort, 2026
- GCI World April 2026 — Data Science, Matsuo-Iwasawa Laboratory, Graduate School of Engineering, The University of Tokyo, 2026

10. LANGUAGES

Amharic — Native English — Fluent

11. REFERENCES

Available upon request